LABORATORY & VIRTUAL TASK REPORT

DNA Isolation & Purification

A Hybrid Virtual Simulation & Home-Based Practical Analysis of Octoploid Fragaria × ananassa Genomes

BIOTECHNOLOGY INTERNSHIP • TASK 3 • CODEALPHA

Theoretical Foundations

Understanding the molecular chemistry of cell wall digestion, cellular membrane lysis, lipid dissolution, and phase-separated nucleic acid precipitation.

Selecting the Model Organism

Fragaria × ananassa

Strawberries exhibit an octoploid genome structure ($8n = 56$ chromosomes), yielding a disproportionately massive concentration of DNA per cell.

This structural abundance allows distinct visual strands of precipitated chromatin to aggregate into high-yield, observable white fibrous meshworks without complex scientific lab machinery.

Comparative Standard

While standard diploid cell extraction methods (e.g., human buccal cells in virtual labs) yield sub-visible microgram pellets, strawberries provide a macro-scale outcome.

This acts as an ideal bridge to validate physical home-laboratory protocols against in-silico simulation standards (such as the LabXchange or Learn.Genetics DNA isolation pipelines).

Chemical Mechanics of Extraction

Lysis Stage

Amphipathic sodium dodecyl sulfate (SDS) or mild dishwashing detergents disrupt the lipid bilayer of cell and nuclear membranes, solubilizing lipids and releasing chromatin.

Ionic Shielding

Sodium chloride ($NaCl$) releases positive sodium ions (Na^+) which shield negatively charged phosphate backbones (PO_4^{3-}) of DNA, encouraging polymer aggregation.

Precipitation

Ice-cold ethanol reduces the dielectric constant of the solvent. Positively charged sodium ions pair with exposed phosphates, rendering DNA insoluble and forcing it to precipitate at the interphase.

Step-by-Step Lab Protocol

1. Pulverization

Mechanical homogenization of strawberry fruit inside a sealed reservoir to break cellulose-rich cell walls.

2. Buffer Lysis

Mixing mechanical pulp with lysis solution (detergent-salt blend) to liberate DNA from nuclear envelopes.

3. Filtration

Gravity-fed paper filtration of coarse homogenate to separate insoluble pectin, skin, and seeds from nucleic acids.

4. Precipitation

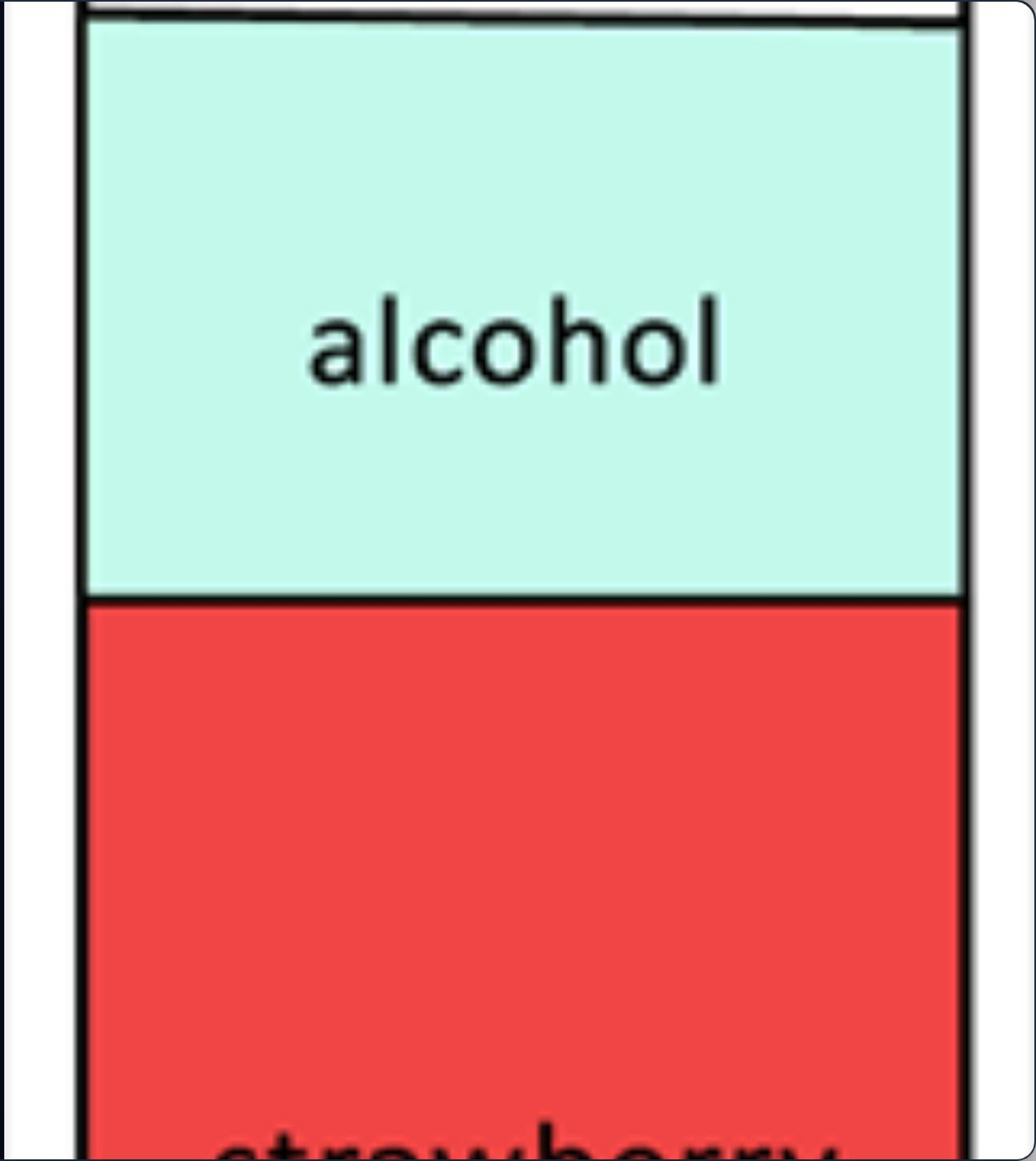
Slowly layering ice-cold absolute ethanol onto filtrate, prompting instantaneous spooling of genomic strands.

Lysis & Cellular Digestion

Mechanical Wall Disruption

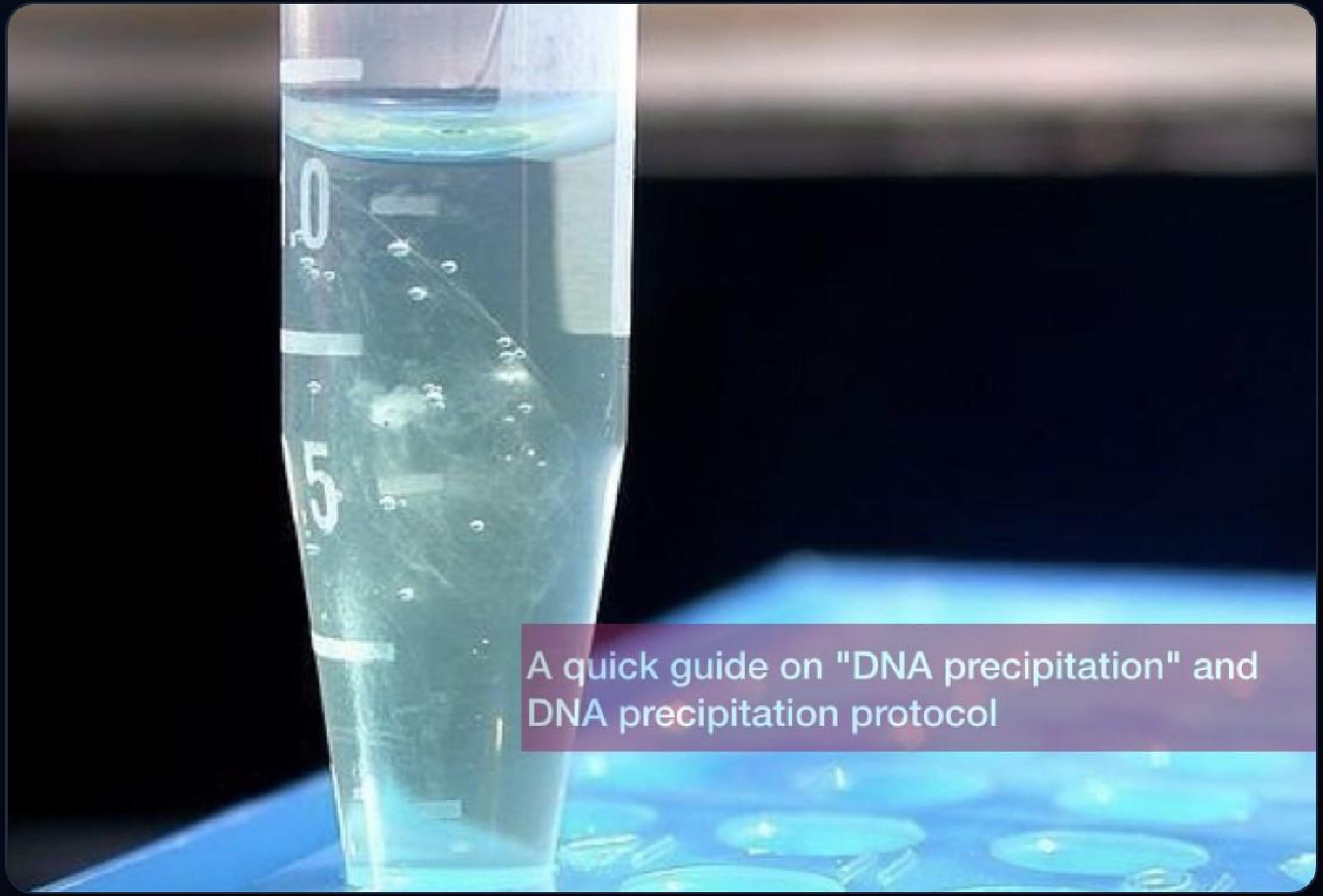
Before chemical extraction begins, structural cellulose cell walls must be physically cleaved.

Mechanical mashing in a polymer sleeve exposes cell walls to high-throughput lysis buffer, maximizing enzyme inhibition and exposing nuclear chromatin payloads.



Isolating Precipitated DNA

- **Ethanol Layering:** Direct addition of ice-cold ethanol creates two distinct aqueous-organic liquid layers.
- **Interphase Emergence:** An immediate formation of transparent, white, stringy web structures occurs at the boundary.
- **Physical Spooling:** Tangled genomic fibers, bundled with nucleosomes, are easily collected onto a sterile glass stirring rod.
- **Impurities Separation:** Pectin and lighter proteins remain fully dissolved in the lower blue-tinted aqueous filtrate.



Simulated Purity Diagnostics

1.84

A260 / A280 Purity Ratio

Ultraviolet Spectrophotometry

To evaluate extraction success without physical lab bio-analyzers, we simulated UV-Vis absorbance profiles of isolated genomic DNA samples.

An A_{260} / A_{280} ratio of 1.80 to 1.90 signifies clean, uncontaminated nucleic acid. Values beneath 1.7 suggest protein impurities, whereas values above 2.0 suggest co-purified RNA residues.

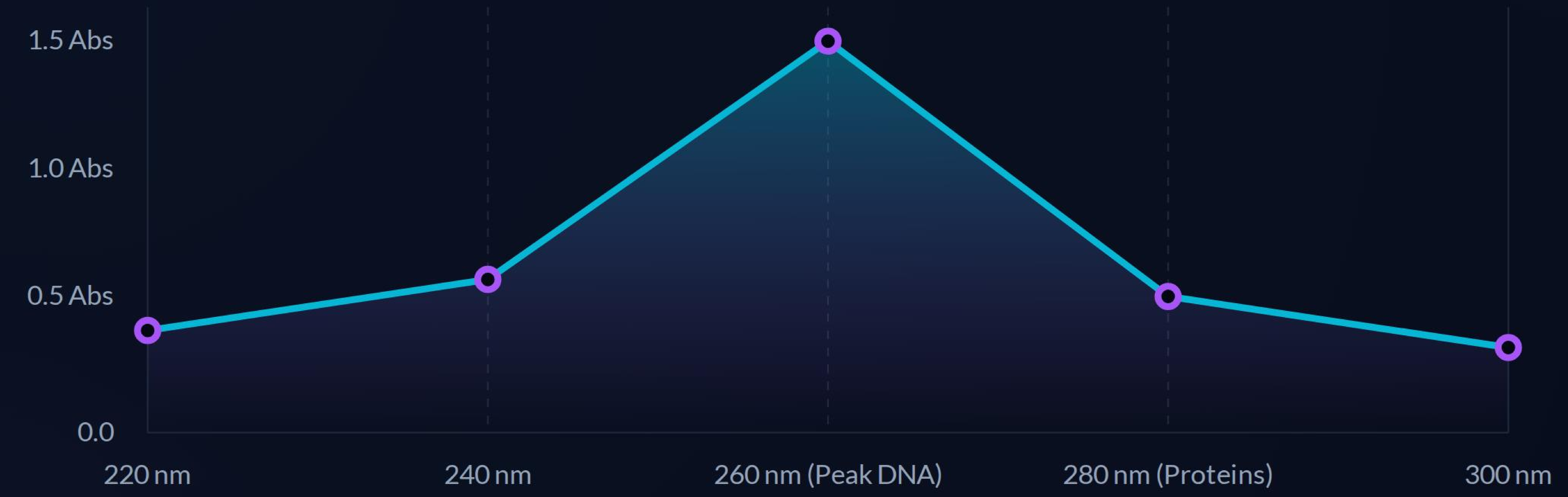
Comparative Analysis Matrix

Fruit Organism	Genetic Ploidy Level	Cellular Disruption Ease	Filtrate Viscosity	Relative DNA Yield
Strawberry (Fragaria)	Octoploid ($8n$)	Very High (Thin cell walls)	Low (High pectinase)	Excellent (Visibly dense bands)
Banana (Musa)	Triploid ($3n$)	Moderate (Dense starch)	High (Thick pulp)	Good (Medium cloudy fibers)
Onion (Allium cepa)	Diploid ($2n$)	Low (Sulfur protein sheets)	Low-Medium	Fair (Highly transparent strings)
Buccal Cell Control (Virtual)	Diploid ($2n$)	High (No cell walls)	Very Low	Microscopic (Requires centrifugation)

Purity Profile Across Trials

UV-spectrophotometry comparison verifies that omitting sodium chloride drastically lowers purity (1.42), as unshielded histone proteins co-precipitate with nucleic acid polymers.

DNA Absorbance Spectrum



Absorbance spectrum validation curve displaying the classic nucleic acid peak at 260 nm. The ratio of absorbance values at 260nm and 280nm ($A_{260} / A_{280} \approx 1.84$*) confirms highly purified molecular extraction.*

Laboratory Verification

This concludes Task 3: Laboratory / Practical DNA Isolation Project.

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Image Sources

Strawberry Extraction pulp and mechanical lysis

http://googleusercontent.com/image_collection/image_retrieval/11087269274048865590_0

White Precipitated DNA Spooling at aqueous/organic interface

http://googleusercontent.com/image_collection/image_retrieval/14853498408698821832_0

Image Sources

https://www.biologycorner.com/resources/DNA-test-tube.png

Source: www.biologycorner.com

https://geneticeducation.co.in/wp-content/uploads/2019/04/DNA-precipitation-.007-e1555499479596.jpeg

Source: geneticeducation.co.in